viewed as an agency for providing a service or fulfilling a specific need, the profit incentive changes; income is generated by 'optimising the utilisation of goods rather than the production of goods'.[29] The notion of thinking about a product in terms of the function it performs, is a logical extension of total quality management philosophy. The aim of total quality management is to satisfy *customer needs*. Customers usually don't care how their needs are met, as long as they are indeed met. Therefore, it should not matter whether a customer's requirements are satisfied by a specific product, or by a service performed in lieu of that product. Holistic thinking therefore offers the possibility of reducing resource consumption rates while still meeting the needs of consumers.[30]

LOOKING AHEAD

Given the complexity of the environmental problems we are facing, it is unlikely that we will be able to discern the long term implications of the decisions we make now. This is particularly true of our technological decisions. For example, there was never a clearly articulated societal goal to become reliant on fossil fuels; this reliance came about because petroleum was able to satisfy specific technical and economic constraints that emerged at particular points in time. In light of the environmental and national security concerns associated with fossil fuel dependence, this choice of a primary energy source now seems to have been less than optimal. This example seems to follow a more general pattern of technology evolution. Recent work provides intriguing evidence that once a particular technology path is chosen, the choice may become 'locked in', regardless of the advantages of the alternatives.[31]

All technological trajectories are shaped by a variety of economic, social, and political forces. They usually cannot be changed without encountering opposition from well entrenched interests (which try to maintain status quo). Sorting out these conflicting interests requires the articulation of broad social goals set by the political leaders, and historically has been achieved only in times of crisis. Thus, harnessing and channeling technology in productive and ecologically sound ways will no doubt prove